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SUMMARY

Software engineer with experience in backend development and data pipeline optimization. Interested in software security. Open to full-time or working student opportunities.

EDUCATION

University of Stuttgart

Stuttgart, Germany

M.Sc. Information Technology (Specialization: Software Engineering)

Oct 2024 – Present

Relevant Coursework:

- **Deep Learning:** Evaluated Knowledge Graph Foundation Models for zero-shot reasoning on relational databases.
- **Distributed Systems:** Built a P2P chat system with leader election, fault detection, and causal ordering.
- **Cloud Computing:** Designed a serverless AWS architecture for an online retail platform (Lambda, DynamoDB, API Gateway, SQS, EventBridge).

Multimedia University

Cyberjaya, Malaysia

B.Eng. (Hons.) Computer Engineering (Minors: Electronics)

Jul 2020 – Oct 2024

CGPA: 3.7/4.0 (First Class Honors).

WORK EXPERIENCE

University of Stuttgart – Institute of Space Systems (IRS)

Stuttgart, Germany

Research Assistant - SOFIA Data Center

Apr 2026 – Present

- Analyzing and optimizing a Python-based data processing pipeline for the SOFIA infrared astronomy archive, reviewing legacy scientific code to identify memory bottlenecks.

Fraunhofer IOSB

Karlsruhe, Germany

Research Assistant

Apr 2025 – Dec 2025

- Developed Java and Spring Boot backend for LLM-based data extraction evaluating 6 models in Asset Administration Shell (AAS) digital twin applications.
- Implemented automated end-to-end tests and prepared 30+ test datasets for model evaluation.

AirAsia

Kuala Lumpur, Malaysia

Software Engineer Intern

Jul 2023 - Oct 2023

- Debugged 14 software bugs and implemented 5 checkout calculation features, enhancing app stability for 10,000+ users.
- Refactored 2 app pages and integrated GraphQL APIs for bilingual content management in Agile sprints.

PROJECTS

Privacy in AI Tools: How Much Do They See?

Apr 2026 – Present

- Researching data exposure in LLM-integrated productivity tools by intercepting and analyzing outbound network traffic using Python and mitmproxy under controlled conditions.

MangoVision

Nov 2023 – Jul 2024

- Developed a deep learning model for mango fruit detection from aerial images by collecting and preprocessing multiple datasets and evaluating deep learning models, achieving 98.5% accuracy (Best Research Project Award, Gold Medal).

SKILLS

Technical Skills:

- **Programming Languages:** Python, Java, C, C++, SQL, JavaScript.
- **AI & Data Science:** PyTorch, Detectron2, scikit-learn, pandas, NumPy, Foundation Models.
- **Tools:** Docker, Git, Spring Boot, Flutter, REST APIs, GraphQL, Insomnia.
- **Cloud Platforms:** AWS, Firebase.
- **Project Management:** Jira, Confluence, Agile/Scrum.
- **Operating Systems:** Linux (Kali, Ubuntu, Pop), Windows.
- **Security Tools:** Burp Suite, Nmap, SIEM(Splunk).

Languages: English (C1), German (B1).